



Josiel Mendonça Soares de Souza

Work permit: Brazilian | **Date of birth:** 18/01/1993 |

Place of birth: Rio Grande do Norte, Brazil | **Nationality:** Brazilian | **Gender:** Male |

Email address: josielsouza@if.ufrj.br

ABOUT ME

Josiel Mendonça is a postdoctoral fellow in Physics at the Federal University of Espírito Santo (Brazil), funded by FAPES, since 2025. He received his Ph.D. in Physics in 2023 from the Federal University of Rio Grande do Norte, with a specialization in Gravitation and Cosmology and a focus on Gravitational Waves.

He is a member of the Einstein Telescope Observational Science Board, the Brazilian Einstein Telescope Synergy (BETS), and collaborates on gravitational-wave research with the Brazilian Center for Physical Research and the National Institute for Space Research.

His research interests lie in Cosmology and the Astrophysics of Gravitational Waves, with a strong emphasis on numerical modeling, data analysis, and parameter estimation. He has extensive experience in Python and C programming, high-performance computing (HPC), and large-scale numerical simulations. He is the developer of **GWDALI**, an open-source Python package available on PyPI, which implements the Derivative Approximation for Likelihoods method for parameter estimation of gravitational-wave sources in user-defined detector networks.

EDUCATION AND TRAINING

01/10/2025 – CURRENT Espírito Santo, Brazil

POSTDOCTORAL FELLOW PPG-Cosmo, Universidade Federal do Espírito Santo

Field of study Gravitation & Gravitational Waves

05/01/2024 – 30/09/2025 Rio de Janeiro, Brazil

POSTDOCTORAL FELLOW Instituto de Física, Universidade Federal do Rio de Janeiro

Field of study Gravitational Waves

01/03/2019 – 12/07/2023 Rio Grande do Norte, Brazil

PHD IN PHYSICS (COSMOLOGY) Universidade Federal do Rio Grande do Norte

Thesis Late-Time Cosmology with Third Generation Gravitational Wave Observatories

Link <https://repositorio.ufrn.br/items/555096fe-221c-4899-83b4-45378abf5fd6>

01/03/2017 – 26/02/2019 Rio Grande do Norte, Brazil

MASTER IN PHYSICS (COSMOLOGY) Universidade Federal do Rio Grande do Norte

Thesis Discriminating cosmological expansion models with third generation gravitational wave detectors

Link <https://repositorio.ufrn.br/items/67ca986a-c3a9-4d87-9765-054534a30908>

01/02/2012 – 17/12/2017 Rio Grande do Norte, Brazil

BACHELOR IN PHYSICS Universidade Federal do Rio Grande do Norte

15/03/2026 – 28/03/2026 Pisa, Italy

TECHNICAL VISIT FUNDED BY ACME Istituto Nazionale di Física Nucleare - Sezione di Pisa

Field of study Physics

LANGUAGE SKILLS

Mother tongue(s): **PORTUGUESE**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
ITALIAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

WORK EXPERIENCE

PHYSICS TEACHER SECONDARY SCHOOL – PHYSICS TEACHER – 01/07/2023 – 31/12/2023 – ESPÍRITO SANTO, BRAZIL

PUBLICATIONS

2025
[On the use of the Derivative Approximation for Likelihoods for Gravitational Wave Inference](#)

de Souza, J. M. S., & Quartin, M. (2025). arXiv preprint arXiv:2510.16955.

Authors: Josiel Mendonça Soares de Souza, and Miguel Quartin | Journal Name: arXiv

2025
[The Science of the Einstein Telescope](#)

Adrian Abac et al JCAP03(2026)081

Authors: Abac, Adrian, et al. | Journal Name: JCAP | Volume, Issue and Pages: 03(2026)081 | Publisher: IOP

2023
[GWDALI: A Fisher-matrix based software for gravitational wave parameter-estimation beyond Gaussian approximation](#)

de Souza, Josiel Mendonça Soares, and Riccardo Sturani. "GWDALI: A Fisher-matrix based software for gravitational wave parameter-estimation beyond Gaussian approximation." Astronomy and Computing 45 (2023): 100759.

Authors: Josiel Mendonça Soares de Souza, Riccardo Sturani | Journal Name: Astronomy and Computing | Volume, Issue and Pages: 45 (2023): 100759 | Publisher: Elsevier

2023
[Luminosity distance uncertainties from gravitational wave detections of binary neutron stars by third generation observatories](#)

de Souza, Josiel Mendonca Soares, and Riccardo Sturani. "Luminosity distance uncertainties from gravitational wave detections of binary neutron stars by third generation observatories." Physical Review D 108.4 (2023): 043027.

Authors: Josiel Mendonça Soares de Souza, Riccardo Sturnai | Journal Name: Physical Review D | Volume, Issue and Pages: 108.4 (2023): 043027 | Publisher: APS: Physical Review Journals

2022
[Cosmography with standard sirens from the Einstein Telescope](#)

de Souza, Josiel Mendonça Soares, Riccardo Sturani, and Jailson Alcaniz. "Cosmography with standard sirens from the Einstein Telescope." Journal of Cosmology and Astroparticle Physics 2022.03 (2022): 025.

Authors: Josiel Mendonça Soares de Souza, Riccardo Sturani, Jailson Alcaniz | Journal Name: Journal of Cosmology and Astroparticle Physics | Volume, Issue and Pages: 2022.03 (2022): 025 | Publisher: IOP Science

2021
[Cosmological model selection from standard siren detections by third-generation gravitational wave observatories.](#)

de Souza, Josiel Mendonça Soares, and Riccardo Sturani. "Cosmological model selection from standard siren detections by third-generation gravitational wave observatories." Physics of the Dark Universe 32 (2021): 100830.

● **NETWORKS AND MEMBERSHIPS**

18/06/2022 – CURRENT Europe
Einstein Telescope Observational Science Board (ET-OSB)

- Member of Divisions:
- Div 1: Fundamental Physics
 - Div 2: Cosmology
 - Div 9: Common Tools

16/01/2024 – CURRENT Brazil
Brazilian Einstein Telescope Science (BETS)

Brazilian Unit of Einstein Telescope Observational Science Board (ET-OSB).
Contribution: Improvements on DALI approximation for Gravitational Wave Data Analysis and Cosmology.

● **CONFERENCES AND SEMINARS**

15/09/2025 – 19/09/2025 São Paulo, Brazil
First International Latin American Conference on Gravitational Waves

Oral Communication: Exploring the DALI Approximation in Gravitational Wave Data Analysis

2025 Rio de Janeiro, Brazil
Seminar: From Fisher to DALI: Advancing Gravitational-Wave Data Analysis

Seminar on Graduate Program in Physics (UERJ, Rio de Janeiro)

2025 São Paulo, Brazil
V Joint ICTP-Trieste/ICTP-SAIFR School on Cosmology

Poster: Beyond Fisher Approach for Gravitational Waves and Cosmology

2024 Espírito Santo, Brazil
III Workshop Antônio Brasil Batista

Oral Communication: GWDALI: Derivative Approximation for Gravitational Wave Likelihoods.

2024 Germany (Remote)
Seminar: Derivative Approximation for Gravitational Wave Likelihoods

Seminar on Amendola-Wetterich cosmology group of the Institute for Theoretical Physics (Heidelberg, Germany)

2024 Lisbon, Portugal
Seminar: GWDALI: Derivative Approximation for Gravitational Wave Likelihoods

Seminar on Center for Astrophysics and Gravitation (Instituto Superior Técnico, Lisbon)

2024 Espírito Santo, Brazil
VII José Plínio Baptista School of Cosmology

Oral Communication: Beyond Gaussianity for Gravitational Wave Data Analysis at high-z Cosmology

2024 Rio de Janeiro, Brazil
XIX Brazilian School of Cosmology and Gravitation

Poster: Derivative Approximation for Likelihoods on Gravitational Waves and Cosmology Using Autodiff.

2024 Bahia, Brazil
VII CosmoSul

Oral Communication: On the Derivative Approximation for Likelihoods of Standard Sirens

2023 São Paulo, Brazil

São Paulo Advanced School on Multi-Messenger Astrophysics

Oral Communication: GWDALI: a Fisher Matrix Based Software to Estimate Standard Sirens Parameters Beyond Gaussianity-Level

2022 Australia (Remote)

Gravitational Wave Physics and Astronomy Workshop (GWPAW)

Poster: On the Distance Measurements of Bright Standard Sirens with 3G GW Detectors

2022 São Paulo, Brazil

Workshop on Classical Gravity and Applications, ICTP-SAIFR

Oral Communication: Late-Time Cosmology With Bright Standard Sirens

2022 Rio de Janeiro, Brazil

25th annual International Conference on Particle Physics and Cosmology (COSMO'22)

Oral Communication: Cosmology With Bright Standard Sirens

2021 Australia (Remote)

14th Edoardo Amaldi Conference on Gravitational Waves

Oral Communication: Forecasts on Cosmography With Gravitational Waves Using Einstein Telescope

2021 Rio Grande do Norte, Brazil

Semana do PPGF-UFRN

Oral Communication: Gravitational Waves Cosmology

2021 Brazil (Remote)

XXXV Encontro de Física do Norte e Nordeste

Oral Communication: Cosmologia das Ondas Gravitacionais

2020 Rio Grande do Norte, Brazil

Semana do PPGF-UFRN

Oral Communication: Probing Dark Energy With Gravitational Waves

2018 Espírito Santo, Brazil

IV José Plínio Baptista School on Cosmology

Oral Communication: Discriminating Cosmological Expansion Models With Third Generation GW Detectors.

HONOURS AND AWARDS

2024

Honorable Mention on 2024 ICTP-SAIFR PhD Prize in Classical Gravity and Applications – ICTP South American Institute for Fundamental Research

2021

Honorable Mention in the event "Semana do PPGF 2021" – Programa de Pós Graduação em Física da Universidade Federal do Rio Grande do Norte

2020

Honorable Mention in the event "Semana do PPGF 2020" – Programa de Pós Graduação em Física da Universidade Federal do Rio Grande do Norte
